

CSES – Mathematics Problem Reference

All tasks: 1 s, 512 MB. Arithmetic is modulo $10^9 + 7$ only where explicitly stated.

1079 – Binomial Coefficients

Compute n values $\binom{a}{b}$ modulo $10^9 + 7$, where $0 \leq b \leq a$. **In:** n ; then n lines **a b**

Out: each $\binom{a}{b} \bmod 10^9 + 7$

Constraints: $n \leq 10^5$; $a \leq 10^6$.

1081 – Common Divisors

Given n positive integers, find two array elements whose greatest common divisor is as large as possible. **In:** n ; **x_1**

... **x_n**

Out: maximum possible gcd

Constraints: $2 \leq n \leq 2 \cdot 10^5$; $x_i \leq 10^6$.

1082 – Sum of Divisors

Let $\sigma(i)$ be the sum of divisors of i . Compute $\sum_{i=1}^n \sigma(i) \bmod 10^9 + 7$. **In:** n

Out: the sum modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^{12}$.

1095 – Exponentiation

For each query, compute $a^b \bmod 10^9 + 7$. By definition here, $0^0 = 1$. **In:** n ; then n lines **a b**

Out: one modular power per query

Constraints: $n \leq 2 \cdot 10^5$; $0 \leq a, b \leq 10^9$.

1096 – Throwing Dice

Count ordered sequences of dice throws, each in $\{1, \dots, 6\}$, whose total sum is exactly n . **In:** n

Out: number of ways modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^{18}$.

1098 – Nim Game II

There are n heaps. A move chooses one nonempty heap and removes 1, 2, or 3 sticks. Last move wins; determine the winner under optimal play. **In:** t ; for each test: n ; heap sizes

Out: **first** or **second** for each test

Constraints: $t, n \leq 2 \cdot 10^5$; $x_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

1099 – Stair Game

Stairs $1 \dots n$ contain balls. A move chooses $k \neq 1$ with balls and moves any positive number of balls from stair k to $k - 1$. Last move wins; with no legal move initially, second wins.

In: t ; for each test: n ; **p_1** **...** **p_n**

Out: **first** or **second**

Constraints: $t, n \leq 2 \cdot 10^5$; $0 \leq p_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

1712 – Exponentiation II

For each query, compute $a^{b^c} \bmod 10^9 + 7$. By definition here, $0^0 = 1$. **In:** n ; then n lines **a b c**

Out: one answer per query

Constraints: $n \leq 10^5$; $0 \leq a, b, c \leq 10^9$.

1713 – Counting Divisors

For each given integer x , report the number of positive divisors of x . **In:** n ; then n values x

Out: divisor count for each x

Constraints: $n \leq 10^5$; $x \leq 10^6$.

1715 – Creating Strings II

Given a lowercase string, count the number of distinct strings obtainable by permuting all of its characters. **In:** one lowercase string of length n

Out: count modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^6$.

1716 – Distributing Apples

Distribute m identical apples among n children; each child may receive any nonnegative number. Count all distributions.

In: n m

Out: number of distributions modulo $10^9 + 7$

Constraints: $1 \leq n, m \leq 10^6$.

1717 – Christmas Party

n children each brought one gift. Count assignments of gifts to children such that nobody receives their own gift. **In:** n

Out: number of valid distributions modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^6$.

1722 – Fibonacci Numbers

With $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-2} + F_{n-1}$, compute F_n . **In:** n

Out: $F_n \bmod 10^9 + 7$

Constraints: $0 \leq n \leq 10^{18}$.

1723 – Graph Paths I

Given a directed graph, count paths from node 1 to node n that use exactly k edges. **In:** n m k ; then m directed edges **a b**

Out: number of length- k paths modulo $10^9 + 7$

Constraints: $n \leq 100$; $m \leq n(n - 1)$; $k \leq 10^9$.

1724 – Graph Paths II

Given a directed weighted graph, find the minimum total weight of a path from node 1 to node n using exactly k edges.

In: n m k ; then edges **a b c**

Out: minimum length, or -1 if no such path exists

Constraints: $n \leq 100$; $m \leq n(n - 1)$; $k \leq 10^9$; $c \leq 10^9$.

1725 – Dice Probability

Throw a fair six-sided die n times. Find the probability that the total sum lies in $[a, b]$. **In:** n **a b**

Out: probability rounded to 6 decimals

Constraints: $n \leq 100$; $1 \leq a \leq b \leq 6n$.

1726 – Moving Robots

Initially every square of an 8×8 board has one robot. Each robot independently makes k random legal orthogonal moves, choosing uniformly among directions that stay on the board. Find the expected number of empty squares afterward. **In:** k

Out: expected empty squares, rounded to 6 decimals

Constraints: $1 \leq k \leq 100$.

1727 – Candy Lottery

n children independently receive a uniformly random integer number of candies from 1 to k . Find the expected maximum received by any child. **In:** n k

Out: expected maximum, rounded to 6 decimals

Constraints: $1 \leq n, k \leq 100$.

1728 – Inversion Probability

For each position i , choose x_i independently and uniformly from $1 \dots r_i$. Find the expected number of inversions (a, b) with $a < b$ and $x_a > x_b$. **In:** n ; **r_1** **...** **r_n**

Out: expected inversions, rounded to 6 decimals

Constraints: $n \leq 100$; $r_i \leq 100$.

1729 – Stick Game

A heap has s sticks. Allowed removals are the distinct values in P , and $1 \in P$. For every initial size $s = 1, \dots, n$, classify the position as winning or losing under optimal play; last move wins. **In:** n k ; $p_1 \dots p_k$
Out: length- n string of W/L
Constraints: $n \leq 10^6$; $k \leq 100$; $p_i \leq n$.

1730 – Nim Game I

There are n heaps. A move chooses one nonempty heap and removes any positive number of sticks. Last move wins; determine the winner under optimal play. **In:** t ; for each test: n ; heap sizes
Out: first or second
Constraints: $t, n \leq 2 \cdot 10^5$; $x_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

2064 – Bracket Sequences I

Count valid parenthesis sequences of total length n . **In:** n
Out: number of sequences modulo $10^9 + 7$
Constraints: $1 \leq n \leq 10^6$.

2164 – Josephus Queries

Children $1 \dots n$ stand in a circle; repeatedly remove every second remaining child until none remain. For each query (n, k) , report the k -th child removed. **In:** q ; then q lines n k
Out: k -th removed child for each query
Constraints: $q \leq 10^5$; $1 \leq k \leq n \leq 10^9$.

2182 – Divisor Analysis

A positive integer is given by its prime factorization $\prod x_i^{k_i}$, with distinct primes x_i . Compute the number, sum, and product of all positive divisors. **In:** n ; then n lines x k
Out: divisor count, divisor sum, divisor product; all modulo $10^9 + 7$
Constraints: $n \leq 10^5$; $x \leq 10^6$; $k \leq 10^9$.

2185 – Prime Multiples

Given k distinct primes a_i and integer n , count integers in $1 \dots n$ divisible by at least one a_i . **In:** n k ; $a_1 \dots a_k$
Out: number of divisible integers
Constraints: $n \leq 10^{18}$; $k \leq 20$; $2 \leq a_i \leq n$.

2187 – Bracket Sequences II

A prefix of a bracket sequence is fixed. Count valid bracket sequences of total length n that begin with exactly this prefix. **In:** n ; prefix string of length k
Out: number of completions modulo $10^9 + 7$
Constraints: $1 \leq k \leq n \leq 10^6$.

2207 – Grundy's Game

Start with one heap of n coins. A move chooses a heap and splits it into two nonempty heaps of different sizes. Last move wins; determine the winner under optimal play. **In:** t ; then t values n
Out: first or second for each test
Constraints: $t \leq 10^5$; $n \leq 10^6$.

2208 – Another Game

There are n nonempty heaps. A move selects some of the currently nonempty heaps and removes one coin from each selected heap. Last move wins; determine the winner. **In:** t ; for each test: n ; heap sizes
Out: first or second
Constraints: $t, n \leq 2 \cdot 10^5$; $x_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

2209 – Counting Necklaces

Count colorings of a necklace with n pearls and m possible colors, where colorings equivalent by rotation are considered the same. **In:** n m
Out: number of distinct necklaces modulo $10^9 + 7$
Constraints: $1 \leq n, m \leq 10^6$.

2210 – Counting Grids

Count black/white colorings of an $n \times n$ grid, considering two grids equal if one can be rotated to obtain the other. **In:** n
Out: number of distinct grids modulo $10^9 + 7$
Constraints: $1 \leq n \leq 10^9$.

2417 – Counting Coprime Pairs

Given n positive integers, count unordered index pairs whose values are coprime, i.e. have gcd 1. **In:** n ; $x_1 \dots x_n$
Out: number of coprime pairs
Constraints: $n \leq 10^5$; $x_i \leq 10^6$.

3154 – System of Linear Equations

Solve n linear equations in m variables over the field modulo $10^9 + 7$: $\sum_j a_{i,j} x_j \equiv b_i \pmod{10^9 + 7}$. Any valid solution is accepted. **In:** n m ; then n rows containing m coefficients and b_i
Out: $x_1, \dots, x_m$ with $0 \leq x_i < 10^9 + 7$, or -1 if impossible
Constraints: $n, m \leq 500$; all coefficients and b_i are in $[0, 10^9 + 7)$.

3355 – Sum of Four Squares

For each nonnegative integer n , find nonnegative integers a, b, c, d such that $n = a^2 + b^2 + c^2 + d^2$. **In:** t ; then t values n
Out: one valid quadruple per test
Constraints: $t \leq 1000$; $n \leq 10^7$; sum of all $n \leq 10^7$.

3396 – Next Prime

For each positive integer n , find the smallest prime strictly greater than n . **In:** t ; then t values n
Out: next prime after each n
Constraints: $t \leq 20$; $n \leq 10^{12}$.

3397 – Permutation Order

Let $p(n, k)$ be the k -th permutation of $1 \dots n$ in lexicographic order. Support both directions: given (n, k) output $p(n, k)$, or given n and the permutation output its rank k . **In:** t ; each test is 1 n k or 2 n $p_1 \dots p_n$
Out: the requested permutation or rank
Constraints: $t \leq 1000$; $n \leq 20$; $1 \leq k \leq n!$.

3398 – Permutation Rounds

Start from sorted array $[1, 2, \dots, n]$. In each round, the element currently at position i moves to position p_i . Find the first positive number of rounds after which the array is sorted again. **In:** n ; permutation $p_1 \dots p_n$
Out: number of rounds modulo $10^9 + 7$
Constraints: $1 \leq n \leq 2 \cdot 10^5$.

3406 – Triangle Number Sums

Triangle numbers are $1 + 2 + \dots + k$ for $k \geq 1$. For each positive integer n , find the minimum number of triangle numbers whose sum is n . **In:** t ; then t values n
Out: minimum count for each test
Constraints: $t \leq 100$; $n \leq 10^{12}$.